IGARSS 2024 - 2024 IEEE International Geoscience and Remote Sensing Symposium | 979-8-3503-6032-5/24/$31.00 ©2024 IEEE | DOI: 10.1109/IGARSS53475.2024.10642101

FOREST FIRE REGIMES CHARACTERIZATION THROUGH HISTORICAL INFORMATION DERIVED FROM MODIS

Soma Satya Praveen Mutyala
National Remote Sensing Centre,
Hyderabad, India

Suraj Reddy R
National Remote Sensing Centre,
Hyderabad, India

Rajashekar G
National Remote Sensing Centre,
Hyderabad, India

ABSTRACT

This study investigates forest fire regimes in India using satellite data, specifically MODIS derived active fire locations and burnt areas (Fire_CCIv5.1). Nine fire regime variables, computed from 2003 to 2020, provide insights into fire incidence, inter-annual variability, seasonality, intensity, fire size distribution, and affected vegetation types across the country. K-Mean clustering was used to identify, eight distinct fire regimes with characteristic attributes. Northwestern India has a preponderance of small agricultural fires, where MODIS cannot detect burned areas. Northeastern India experiences intense, frequent forest fires linked to biomass burning from shifting cultivation, with low inter-annual variability. In central Indian deciduous forests, fires are large, frequent, and have a summer peak season with lower inter-annual variability. These findings on the spatial and temporal characteristics of fire are useful for effective fire management and mitigation planning.

Index Terms— Fire Regime, MODIS, Fire_CCI, K-Means

1. INTRODUCTION

Forest fire is considered as one of the significant disasters that result in adverse ecological, economic, and social impacts on a global scale. According to the 'India State of Forests report [1] , 24.56% of the country's total geographic area is covered by trees and forests, encompassing 16 distinct forest types. Among these, tropical dry deciduous forests dominate, followed by tropical moist deciduous forests are more susceptible to fires. On average, 53% of the country's forest cover is susceptible to fires ranging from heavy to light, with 6.17% facing the risk of severe fire damage. While the fire season and peak fire month vary across the country, fires typically occur between January and June.

Fire regime describes the spatial and temporal characteristics of fires, based on the occurrence, spread, behavior, and effects of fires, and includes incidence, seasonality, intensity, and severity, as well as various fire types and patterns [2]. These patterns arise from complex and dynamic processes caused by the interaction and feedback among factors such as fire, climate, characteristics of vegetation, features of the landscape, land use, and ignition patterns, and vary at multiple temporal and spatial scales [3]. However, the pattern and extent of constraints associated with fire characteristics can be predicted. Studies on the fire regimes of extensive geographical regions largely depend on remotely sensed data. Therefore, the utilization of remote sensing plays a crucial role in delineating fire regimes at both global and local levels.

A national level characterization of fire regimes is crucial for forest fire management and mitigation planning. In this study we have used active forest fire locations and burned area information derived from MODIS to characterize fire regimes in India.

2. DATA AND METHODS

2.1. Datasets

Accurately estimating the shape and size of large fires using active fire data from polar-orbiting satellites is not possible, as it can only identify warm flame fronts during satellite overpasses. Mapping medium and large fires more reliably relies on burned area products derived from remote sensing data. However, for extremely small fires, the minimum detectable burned area size is significantly larger compared to active fire detection, which can identify fires up to 1000 times smaller [4]. In this study, we integrate information from burned areas and active fires to overcome these limitations, utilizing monthly Fire_CCI burnt date maps from Google Earth Engine [5] and MCD14ML active fire data from 2003 to 2020. Additionally, vegetation cover type maps generated by National Remote Sensing Centre at 56m are resampled to 250m resolution are incorporated into the study [6].

2.2. Computation of fire regime variables

We computed nine variables to characterize the fire regimes in India. The variables were chosen to specify spatial, temporal and intensity attributes of fire regimes by taking into account fire incidence, inter-annual variability in fire incidence,

Authorized licensed use limited to: University of Missouri Libraries. Downloaded on March 25,2026 at 21:27:54 UTC from IEEE Xplore. Restrictions apply.

seasonality, intensity, size distribution, and vegetation type affected. All nine variables were calculated on a 0.5 degree grid and are assumed to properly summarize spatiotemporal fire characteristics. The variables used are (1) Mean Annual Fire Density (MAFD); (2) Coefficient of Variation in annual Fire Density (CoVFD); (3) Mean Annual Area Burned (MAAB); (4) Coefficient of Variation in annual Area Burned (CoVAB); (5) Fire Season Duration (FSD); (6) Fire Peak Month (FPM); (7) Fire Radiative Power (FRP); (8) Gini Index (GI); and (9) Major Vegetation Type affected by fire (MVT). The Jenks natural breaks method was selected to segment continuous data for classifying all variables.

For each grid, MAFD & MAAB are the mean values of average annual fire locations and burnt areas respectively. CoVFD & CoVAB are coefficient of variations of average annual fire locations and burnt areas respectively. GI for fire sizes is used to characterize the inequality in fire size distributions. High GI values indicate burned area concentration in a small number of large fires. The individual fires are identified by applying the modified flood-fill algorithm in burned area maps [7]. FSD is defined as the duration of the period starting from when 10% of fire pixels were seen in a grid cell to when 90% of the fire pixels were detected. FPM is the month with the most active fire pixels within a grid. FRP is computed from the mean of fire radiative power information are directly retrieved from active fire data. The percentage of vegetation types affected by fires is calculated as the ratio of total burned pixels in each vegetation type and the total burned pixels in the grid cell.

2.3. Fire regimes identification

The nine fire regime variables were preprocessed by excluding grid cells with $<$10% forest area and normalizing the data prior to applying the clustering algorithm. The k-means clustering algorithm [8] was employed in this work because of its simplicity, computational efficiency. The number of clusters was selected with the "elbow" method in the plot of the within-cluster sum of the squared distances (SSD), the optimum of the Bayesian information criterion (BIC), and a silhouette analysis [9].

3. RESULTS AND DISCUSSION

Fig. 1.Ia and 1.Ib shows the history of forest fire locations and burned area from 2003 to 2020. We can clearly see patterns of active locations and burned areas does not match. Burned area is seemed to follow a systematic cycle, while no such pattern is observed fire locations.

3.1. Cluster analysis for forest fire regimes

The different MODIS datasets present interestingly different results regarding the inter-annual variability of fire incidence. From Figure 1.IIIa and 1.IIIb, it is clearly evident that in north eastern states of India even though the MAFD is very high, the MAAB from these fires is very low. This is majorly attributes to the dominance of controlled fires from shifting cultivation within forests in the north eastern states of India. As expected, a low inter-annual coefficient of variation (in both the annual area burned and the annual active fire density) was observed in these areas where high fire incidence was detected.

In the regions where fire events rarely occur, the fire incidence has been highly varied. High inter-annual variability is observed in both burned area and fire density in the North and Northwestern parts of India (Fig. 1.IIIc & 1.IIId). Overall, the inter-annual variability of fire incidence in most of southern, central and northeastern India is relatively low. The maps of active fires have shown a relatively higher fire densities in dry deciduous forests of central India, however the burned area maps have revealed that the most of the areas under dry and moist deciduous forests are highly prone to forest fires with very less inter-annual variability.

The fire season duration is very short in Northern temperate forest and Northwestern parts of India which is dominated by shrubs and grasslands. The fire season in India varies from January to June with majority of fires occur in summer season mostly in May and June months. Very high fire radiative power is detected mostly in Northeast India where majority of the fires here are controlled fires where large amounts of biomass is burnt which is also indicative from the higher FRP from Fig. 1.IIIf. Central Indian deciduous forests which are having higher annual burned areas have relatively lower FRP compared to Northeast India which are mostly dominated by ground fires with little fuel load. Northern temperate forests and Northwestern regions of India showed minimal FRP.

Gini index is relatively low in Northeastern and Northern India showing that area burned is more evenly distributed throughout many small fires. While most of the deciduous forest of India shows the higher Gini index values indicating that the area burned is concentrated in a small number of large fires. Very low Gini indices were found in the areas where very few fires occurred (Fig. 1.IIIg). Unsurprisingly, in majority of the grids that are affected fires are distributed among the dry and moist deciduous forests of India. In Northeastern India, in majority of the grids evergreen forests are affected, while in Northern and Northwestern parts of India, other vegetation categories are affected.

3.2. Classification of fire regimes and potential drivers

We identified the number of clusters for the k-means clustering algorithm was based on the results of the scree plot showing the SSD values (Fig. 1.IIa), the BIC criterion and a silhouette analysis (Fig. 1.IIb). From Fig 1.IIa the variation of BIC value is small after k=8. The "elbow" criterion applied to the scree plot also suggests that k should not be larger than eight, the selected number of clusters was eight. This prefer-

Authorized licensed use limited to: University of Missouri Libraries. Downloaded on March 25,2026 at 21:27:54 UTC from IEEE Xplore. Restrictions apply.

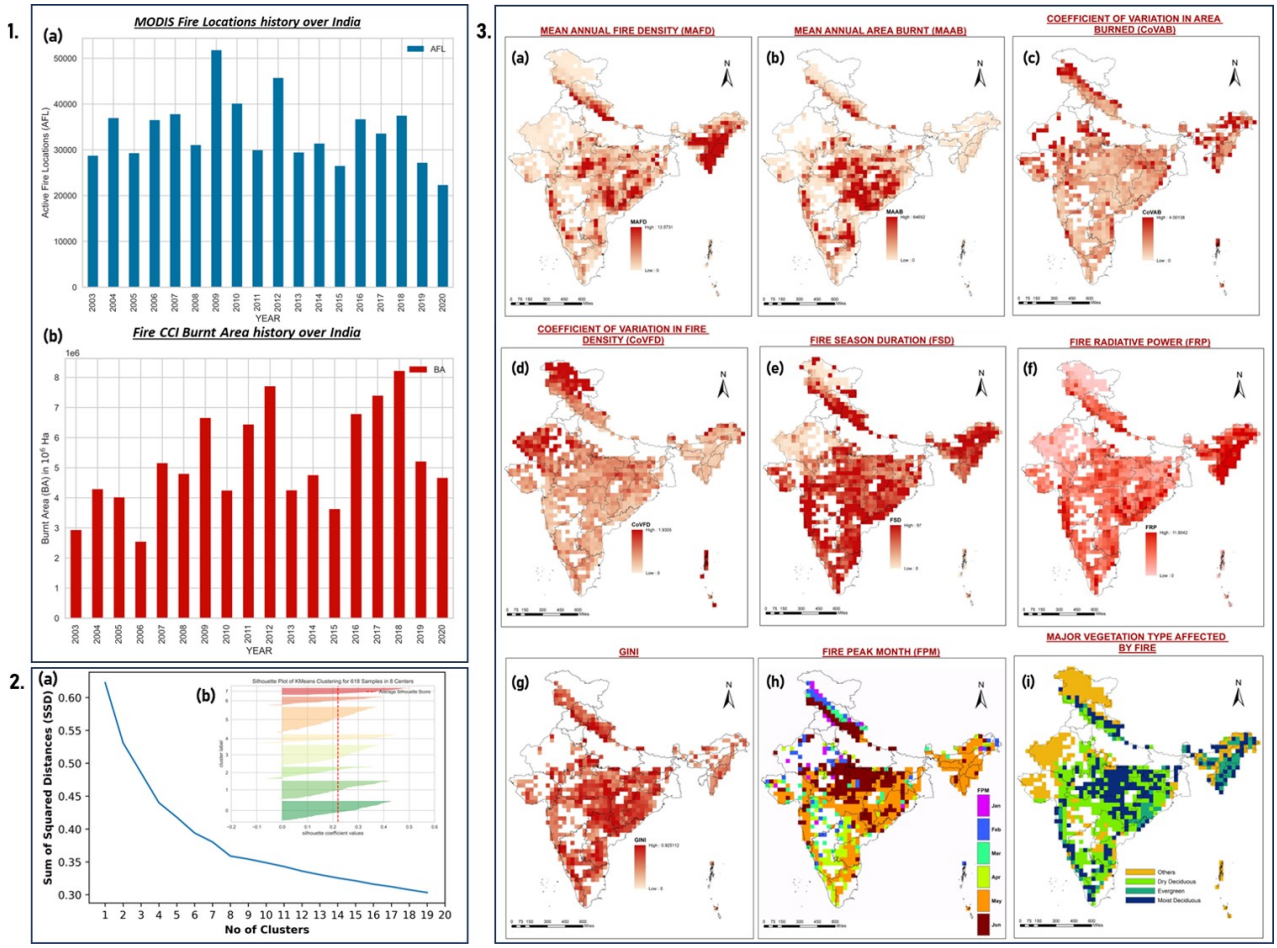

Fig. 1. I(a) and (b) Bar graphs showing Modis active fire locations and Fire_CCI burnt area from 2003 to 2020 for India, II. (a) Scree plot showing with-in class sum of squared distances by number of clusters, (b) silhouette plot for k-means clustering (for k=8), III. Maps showing (a) MAFD; (b) MAAB; (c) CoVAB; (d) CoVAB; (e) FSD; (f) FRP; (g) GI; (f) FPM; and (g) MVT.

able choice was also confirmed by a silhouette criterion and a silhouette analysis (Fig. 1.IIb) out of a range of two to twenty clusters. The eight fire regimes distinguished with respect to central tendency (i.e. median) in fire incidence and its inter-annual variability, seasonality, intensity, and Gini index (Fig. 2b) and this classification led to the labels shown in the last row of Table in Fig.3. Finally, the spatial distribution of fire regimes in India is mapped in Fig. 2a. The eight fire regimes distinguished with the described clustering procedure are detailed below:

1. **HHLLHHH53**: has the fires that are large, frequent and intense with less inter-annual variability. This regime has longer fire season majorly covering moist deciduous forests.

2. **MMLLHHH54**: has moderately frequent intense fires majorly covering other minor vegetation types and experience small number of larger fires.

3. **MLMMMMH52**: has less frequent fire incidences with moderate variability and intensity majorly covering evergreen forests and experience small number of larger fires.

4. **LLMMMML52**: has infrequent evenly sized fires with moderate variability and intensity majorly covering evergreen forests. Also experience shorter fire season.

5. **LHLMHHM53**: has very high frequent intense fires covering larger areas extending over small regions. This regime almost entirely spanning over Northeastern India where shifting cultivation is dominant.

6. **MLMMMMH54**: represents moderately frequent that are covering smaller areas and see small number of larger fires in other minor vegetation types.

7. **HMLMMHH52**: has high frequent fires with moderate intensity. Majorly dominated small number of large fires and experience longer fire season.

Authorized licensed use limited to: University of Missouri Libraries. Downloaded on March 25,2026 at 21:27:54 UTC from IEEE Xplore. Restrictions apply.

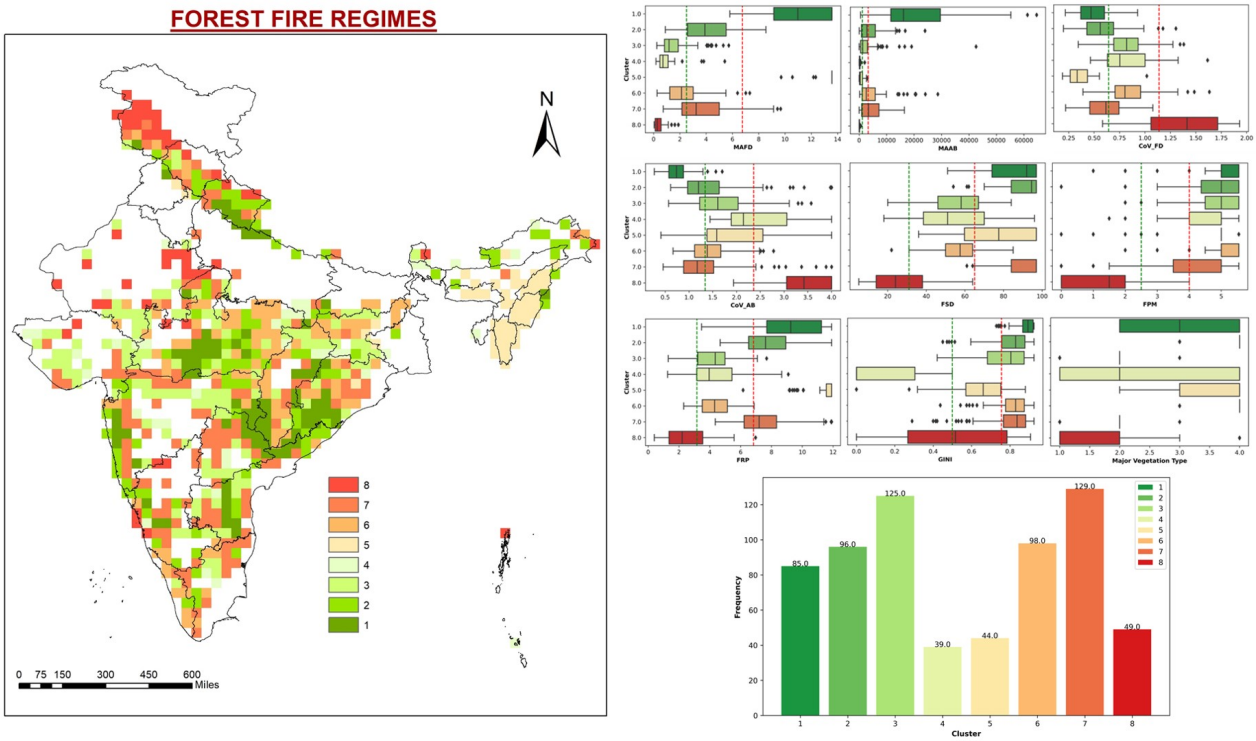

Fig. 2. (a) Forest fire regimes of India; (b) Clusterwise boxplots of each variable (red and greesn horizontal dashed lines are jenks breaks cutoffs to classify as high, medium and low); (c) Frequency distribution of forest fire regimes. (Vegetaiontype classes 1 – Dry Deciduous forests, 2 – Evergreen forests, 3 - Moist Deciduous forests and 4 – Other forests)

	Index of fire regime							
Cluster	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**
MAAB	High(16160)	Medium(2830.5	Medium(1305)	Low(155)	Low(383)	Medium(2602.5)	High(3384)	Low(63)
MAFD	High(10.98)	Medium(3.91)	Low(1.17)	Low(0.71)	High(13.57)	Low(2.13)	Medium(3.25)	Low(0.28)
CoV_FD	Low(0.47)	Low(0.56)	Medium(0.82)	Medium(0.75)	Low(0.33)	Medium(0.8)	Low(0.62)	High(1.41)
CoV_AB	Low(0.74)	Low(1.2)	Medium(1.6)	Medium(2.14)	Medium(1.58)	Medium(1.38)	Medium(1.17)	High(3.42)
FRP	High(9.26)	High(7.61)	Medium(4.36)	Medium(3.96)	High(11.9)	Medium(4.31)	Medium(7.2)	Low(2.21)
FSD	High(92)	High(94.5)	Medium(58)	Medium(51)	High(77.5)	Medium(57.5)	High(97)	Low(24)
GINI	High(0.89)	High(0.83)	High(0.8)	Low(0.1)	Medium(0.66)	High(0.83)	High(0.84)	Medium(0.52)
FPM	5	5	5	5	5	5	5	2
MVT	3	4	2	2	3	4	2	2
Regime	**HHLLHHH53**	**MMLLHHH54**	**MLMMMMH52**	**LLMMMML52**	**LHLMHHM53**	**MLMMMMH54**	**HMLMMHH52**	**LLHHLLM22**

Fig. 3. Fire characteristics and classification of fire regimes; Median to determine the scale of high, medium and low is shown in brackets.

8. **LLHHLLM22**: has relatively low fire incidence, irregular with low intensity and shorter fire season with major fire episodes in February.

4. CONCLUSION

In conclusion, this study provides a comprehensive analysis of vegetation fire regimes in India over the period 2003-2020, utilizing active fire locations and burnt areas derived from MODIS. The investigation reveals distinctive fire regime characteristics, including patterns of incidence, inter-annual variability, seasonality, intensity, fire size distribution, and the affected vegetation types across the country. We clearly identified Northeastern India where intense biomass burning happen as a single regime. Overall majority of the country experiences small number of larger fire episodes. We employed K-Mean clustering to identify eight distinct fire regimes with specific attributes, shedding light on regional variations. The findings contribute valuable insights for effective fire management and mitigation planning at a national scale, emphasizing the importance of remote sensing in understanding and predicting fire regimes for informed decision-making.

Authorized licensed use limited to: University of Missouri Libraries. Downloaded on March 25,2026 at 21:27:54 UTC from IEEE Xplore. Restrictions apply.

5. REFERENCES

[1] Forest Survey of India, "India state of forest report (isfr)," 2021.

[2] Sally Archibald, Caroline E. R. Lehmann, Jose L. Gómez-Dans, and Ross A. Bradstock, "Defining pyromes and global syndromes of fire regimes," *Proceedings of the National Academy of Sciences*, vol. 110, pp. 6442–6447, 4 2013.

[3] Ana M. G. Barros and José M. C. Pereira, "Wildfire selectivity for land cover type: Does size matter?," *PLoS ONE*, vol. 9, pp. e84760, 1 2014.

[4] Louis Giglio, Tatiana Loboda, David P. Roy, Brad Quayle, and Christopher O. Justice, "An active-fire based burned area mapping algorithm for the modis sensor," *Remote Sensing of Environment*, vol. 113, pp. 408–420, 2 2009.

[5] M Padilla Parellada, "Esa fire climate change initiative (fire_cci): Modis fire_cci burned area pixel product, version 5.1. centre for environmental data analysis," 2018.

[6] C. Sudhakar Reddy, C. S. Jha, P. G. Diwakar, and V. K. Dadhwal, "Nationwide classification of forest types of india using remote sensing and gis," *Environmental Monitoring and Assessment*, vol. 187, pp. 777, 12 2015.

[7] S. Archibald and D. P. Roy, "Identifying individual fires from satellite-derived burned area data," 2009, pp. III–160–III–163, IEEE.

[8] Douglas. Steinley, "K-means clustering: A half-century synthesis," *British Journal of Mathematical and Statistical Psychology*, vol. 59, pp. 1–34, 5 2006.

[9] Dongmei Chen, José M.C. Pereira, Andrea Masiero, and Francesco Pirotti, "Mapping fire regimes in china using modis active fire and burned area data," *Applied Geography*, vol. 85, pp. 14–26, 8 2017.

Authorized licensed use limited to: University of Missouri Libraries. Downloaded on March 25,2026 at 21:27:54 UTC from IEEE Xplore. Restrictions apply.